Detecting and Rewriting Gendered Language in Job Postings: A Two-Stage BERT and LLM Pipeline

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Abstract

Research in social psychology has shown that the specific words used in job advertisements influence perceived job appeal and sense of belonging among potential applicants. Gaucher, Friesen, and Kay (2011) demonstrated that masculine-coded wording, terms like "competitive," and "assertive", deters women from applying, while feminine-coded wording has no comparable effect on men. This asymmetry means that masculine language in postings actively narrows applicant pools, yet no automated system exists to detect and correct it at scale. Here, I present a two-stage NLP pipeline that fine-tunes BERT to classify job posting sentences as masculine-coded, feminine-coded, or neutral, and then uses three LLMs (GPT-4o, Claude, Llama 3.3) to rewrite masculine sentences as gender-neutral. Testing the classifier on 10,000 postings from four industries confirms that male-dominated industries use significantly more masculine language (72.2% vs. 60.2% of postings). Evaluation of the rewrites shows that automated and human assessments produce inverted model rankings, demonstrating that both evaluation modes are necessary to maximize rewrite quality. These results suggest that transformer-based detection combined with LLM rewriting offers a viable approach to reducing gendered language in job postings.

1 Introduction

Language used in job postings is not gender-neutral. Gaucher, Friesen, and Kay (2011) found that job advertisements in male-dominated industries contain significantly more masculine-coded words ("Ex: competitive, dominant) than those in female-dominated industries. Notably, exposure to masculine-coded postings reduced women's perceived job appeal and anticipated belongingness, while feminine-coded postings had no comparable effect on men. This asymmetry suggests that masculine language deters women from applying to certain jobs without affecting men's interest.

Despite this finding, practical tools for addressing gendered language remain limited. Gaucher et al. provided word lists for identifying masculine and feminine terms, but keyword matching alone cannot distinguish genuinely gendered usage ("competitive environment") from structural usage ("competitive salary"). To my knowledge, no prior work combines contextual sentence-level detection with automated rewriting and dual evaluation of the results.

This paper presents a two-stage pipeline: (1) a fine-tuned BERT classifier (Devlin et al., 2019) that identifies masculine-coded, feminine-coded, and neutral sentences in job postings, and (2) a rewriting stage in which three LLMs, GPT-4o (OpenAI, 2024), Claude Sonnet 4 (Anthropic, 2024), and Llama 3.3 70B (Meta AI, 2024), rewrite the detected masculine sentences as gender-neutral.

2 Methodolgy

2.1 Data

I use the Kaggle LinkedIn Job Postings dataset (2023–2024), ~124,000 postings with full descriptions, titles, and industry labels. After joining postings with industry mappings, stripping HTML artifacts, and enforcing a 150-word minimum, 50,606 postings remain across four target industries: Tech (16,509), Finance (10,217), Healthcare (20,547), and Education (3,333). These industries were selected to span the gender-segregation spectrum: Tech and Finance are male-dominated, while Healthcare and Education are female-dominated, enabling cross-industry comparison of gendered language patterns.

Each posting was segmented into sentences using NLTK's sent_tokenize. Sentences containing

any word matching the Gaucher et al. (2011) masculine or feminine stems were flagged for annotation. The original lexicon was expanded using Word2Vec embeddings (Mikolov et al., 2013) trained on Google News, adding 3 masculine and 9 feminine terms after manual curation and cross-list filtering. The full expanded lexicon is provided in Appendix C.

Sample	# P	S	Flag	M	F	N
Random	300	6,534	2,226 (34.1%)	332 (14.9%)	426 (19.1%)	1,468 (65.9%)
Enriched	300	13,320	5,745 (43.1%)	1,353 (23.6%)	1,170 (20.4%)	3,057 (53.2%)

Table 1. Annotation summary. P = postings, S = sentences, Flag = flagged by lexicon match. M = masculine (includes collapsed "both" labels), Fem = feminine, N = neutral.

2.2 Annotation

Two sampling strategies produced 300 postings each (75 per industry): a random sample drawn uniformly from the corpus, and an enriched sample drawn from the postings with the highest gendered keyword density. The random sample represents what the classifier will encounter in deployment; the enriched sample provides more keyword-flagged training examples, though higher keyword density does not guarantee more genuinely gendered language since context determines whether a word is used in a gendered way.

Flagged sentences were pre-labeled by Claude Sonnet (Anthropic, 2024) as masculine, feminine, neutral, or both, then reviewed by a human annotator. This semi-automated approach was necessary because keyword matching alone cannot distinguish gendered from neutral usage. For example, "competitive environment" and "competitive salary" contain the stem compet*, but only the former describes workplace culture in a gendered way. Additionally, following Gaucher et al. 's finding that the masculine component of mixed-gender language carries the exclusionary effect, sentences labeled "both" were included in the masculine class, resulting in a final 3-class label set: masculine, feminine, neutral.

2.3 BERT Fine-Tuning

Bert-base-uncased (Devlin et al., 2019) was fine-tuned for 3-class sentence classification using the HuggingFace Transformers library (Wolf et al., 2020). Training used an 80/20 posting-level split that was stratified to ensure balanced industry and gendered label representation across both sets, as well as ensuring each posting is part of only one of the partitions to prevent data leakage.

To address class imbalance (neutral sentences outnumber gendered ones approximately 9:1), I applied inverse-frequency class weights to CrossEntropyLoss. The best checkpoint was selected by validation F1. Hyperparameters followed the ranges recommended by Devlin et al.

Each multiclass model was trained (random and enriched) and evaluated on a combined test set (20% random + 20% enriched). The enriched-trained model achieved the higher macro F1 (0.85 vs. 0.83) and was used for all downstream tasks.

Then, to validate that the classifier captures real sociolinguistic patterns, the enriched model was applied to 10,000 unseen postings (2,500 per industry, totaling 213,779 sentences) not used in training or testing.

2.4 Rewriting Pipeline

Candidate selection: Test-set postings were ranked by their proportion of BERT-predicted masculine sentences, and the top 3 per industry were selected (12 postings total).

Rewriting: Each posting was sent to three LLMs (GPT-4o, Claude Sonnet 4, and Llama 3.3 70B Instruct) with a specific prompt. The full posting was provided as input with [FLAGGED] tags on masculine-classified sentences. The prompt included the gendered word lists and instructed models to convert masculine language to neutral language while leaving semantic meaning intact, and to mark modified sentences with [EDITED] tags. All models ran at temperature 0 for reproducibility. The full prompt is provided in Appendix A.

Automated evaluation: The enriched multiclass model was re-run on both original and rewritten sentences. Success was defined as a masculine-to-neutral or masculine-to-feminine prediction shift on edited sentences. Three postings were excluded from all LLMs due to formatting issues from GPT-4o, leaving 9 postings for evaluation.

Human evaluation: Four raters evaluated 7 items in a blinded comparative ranking survey. Each item showed an original sentence and three rewrites (labeled Model A, B, C). Raters selected which rewrite best removed gendered language and which best preserved meaning. The full survey is provided in Appendix B.

3 Results

3.1 Classification Performance

Tables 2 & 3 show the enriched multiclass model's performance & confusion matrix on the combined test set. A heatmap visualization of the confusion matrix is provided in Appendix D.

Class	Precision	Recall	F1
Masculine	0.75	0.82	0.78
Feminine	0.80	0.83	0.81
Neutral	0.96	0.95	
Macro Avg			0.85
Accuracy			0.93

Table 2: Enriched multiclass model on the combined test set (3,716 sentences).

Predicted	Neutral	Masculine	Feminine
Neutral (3,054)	2,922	77	55
Masculine (345)	50	284	11
Feminine (317)	35	20	262

Table 3: Confusion matrix for the enriched multiclass model (3,716 sentences). Rows are true labels, columns are predicted labels.

3.2 Corpus-Wide Industry Analysis

Industry	% M	Avg M/P	%F	Avg F/P
Finance	77.0%	1.60	53.4%	1.18
Tech	67.4%	1.43	49.2%	1.01
Healthcare	63.9%	1.43	73.3%	1.87
Education	56.6%	1.11	70.6%	1.66

Table 4: Gendered language prevalence (10,000 postings, 2,500/industry). %M/%F = postings with masculine/feminine sentences. Avg M/P, Avg F/P = average per posting.

3.3 LLM Re-write Evaluation

Original: "…designing and delivering advanced commercial capabilities and **ambitious** digital experiences to help the business launch their next big **bets**…"

Claude: "…and **forward-thinking** digital experiences…their next big **bets**…"
GPT-4o: "…and **innovative** digital experiences…their next big **bets**…"
Llama 3.3: "Working **collaboratively**…to **innovate** and build…and **innovative** digital experiences…their next big **initiatives**…"

Figure 1: Example rewrite comparison. Original: Healthcare posting, Director of Events and Experiential Marketing. The masculine-coded word "ambitious" was flagged by BERT.

LLM	BERT SR	Human GN	Human SP
Sonnet 4	76.5%	16.7%	5.6%
Llama 3.3	62.5%	50.0%	33.3%
GPT-4o	53.1%	33.3%	61.1%

Table 5: Automated and human evaluation of LLM rewrites. BERT SR measured on 9 postings/51 sentences. Human win rates = how often each model was selected as best (4 raters, 7 items). SR = success rate, GN = gender neutrality, SP = semantic preservation.

4 Discussion

Training data composition: Training on postings enriched for gendered keywords provided more positive examples without causing overfitting. The enriched sample did not guarantee genuinely gendered language, but increased the likelihood of gendered usage, and the LLM pre-labeling step resolved ambiguous cases.

Classifier error analysis: The confusion matrix (Table 3) reveals that the classifier's primary failure mode is the gendered-vs-neutral boundary, not masculine-vs-feminine confusion. Of 345 masculine sentences, 50 (14.5%) were misclassified as neutral. Similarly, 35 of 317 feminine sentences (11.0%) were missed. These false negatives likely involve ambiguous contexts: "lead" in "lead a cross-functional team" describes a workplace dynamic (masculine-coded), while "lead quarterly reviews" describes a job duty (neutral). Both contain the same Gaucher stem (lead*), and only surrounding context differentiates them. In the other direction, 77 neutral sentences were incorrectly flagged as masculine and 55 as feminine, false positives where the model failed to recognize that a gendered-associated word was being used in a structural or neutral context ("competitive salary" or "independent contractor"). Cross-class confusion between masculine and feminine was rare (11 + 20 cases), meaning the model reliably identifies the direction of gendered language when detected. The main challenge is distinguishing gendered usage (traits, culture, values) from structural usage (compensation, duties, legal terms).

Industry validation: The corpus-wide analysis (Table 4) confirms that Gaucher et al.'s (2011) patterns persist in modern 2023-2024 LinkedIn postings: male-dominated industries use more masculine language (72.2% vs. 60.2%). The model, trained on only 240 postings, generalizes to 10,000 unseen postings, validating both the

annotation method and the learned representations. These results were consistent with expectations based on Gaucher et al.

Rewrite evaluation: BERT reclassification ranked the LLMs as Claude (76.5%) > Llama (62.5%) > GPT-4o (53.1%) on masculine-to-neutral conversion rate. Human raters produced the opposite ranking for gender neutrality: Llama (50.0%) > GPT-4o (33.3%) > Claude (16.7%). The difference likely traces to the rewriting strategy. Claude made minimal edits, often substituting a single word, sufficient to shift the classifier's prediction but not perceived by humans as meaningfully more neutral. Llama 3.3 made broader changes to phrasing and tone that humans preferred, while GPT-4o made targeted substitutions that best preserved meaning. These differences are partly a function of the prompt, which instructed models to replace masculine-coded words with neutral alternatives without adding new information or changing structure. Claude followed these constraints most literally, while Llama interpreted them more freely. The divergence highlights two points: (1) automated reclassification rewards edits that flip the classifier's label regardless of whether the rewrite reads as more neutral to a human, and (2) prompt design directly shapes rewriting behavior, since models interpreted the same constraints differently. Both findings argue for complementary human evaluation alongside automated metrics.

Ethical considerations: Removing masculine-coded language from a posting does not change underlying workplace culture, since a rewritten posting could mask a non-inclusive environment. The pipeline is also limited to English: gendered language operates differently across languages. The pipeline is best positioned as an assistive tool for recruiters, not an automated replacement for broader DEI efforts.

Limitations: The human evaluation included only 4 raters and 7 items, making rewrite results directional rather than statistically significant. Three of 12 postings were excluded due to formatting issues. Sentence-level classification may also miss gendered tone that emerges from cumulative patterns across an entire posting rather than individual sentences.

5 Conclusion

This paper presented an end-to-end pipeline for detecting and rewriting gendered language in job postings. A fine-tuned BERT classifier achieves macro F1 = 0.85 on 3-class sentence classification. Applied to 10,000 LinkedIn postings, it confirms that the gendered language patterns identified by Gaucher et al. (2011) persist in modern data. Three LLMs rewrote masculine-coded sentences with varying strategies. Automated and human evaluation produced inverted model rankings, demonstrating that both evaluation methods are necessary.

Future work should include larger-scale human evaluation for statistically robust model comparison, research into what density of masculine language constitutes meaningful harm at the posting level, and applicant studies measuring whether rewritten postings actually increase application rates among women.

References

Anthropic. (2024). The Claude Model Family. **Anthropic Technical Report**.

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. **Proceedings of NAACL-HLT**, 4171--4186.

Gaucher, D., Friesen, J., & Kay, A. C. (2011). Evidence That Gendered Wording in Job Advertisements Exists and Sustains Gender Inequality. **Journal of Personality and Social Psychology**, 101(1), 109--128.

Meta AI. (2024). The Llama 3 Herd of Models. **Meta AI Technical Report**.

Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient Estimation of Word Representations in Vector Space. **Proceedings of ICLR Workshop**.

OpenAI. (2024). GPT-4 Technical Report. **arXiv preprint arXiv:2303.08774**.

Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., Cistac, P., Rault, T., Louf, R., Funtowicz, M., Davison, J., Shleifer, S., von Platen, P., Ma, C., Jernite, Y., Plu, J., Xu, C., Le Scao, T., Gugger, S., ... Rush, A. M. (2020). Transformers: State-of-the-Art Natural Language Processing. **Proceedings of EMNLP: System Demonstrations**, 38--45.